Kogger Wi-Fi Bridge

ESP32-C3 UART ↔ Wi-Fi module firmware · datasheet

Firmware **0.22.0**
2026-11-05
KOGGER LLC · kogger.tech
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Overview

Firmware for a small ESP32-C3-MINI-1U board that puts serial devices on Wi-Fi. Two UART lines are relayed to UDP ports or TCP clients as **whole protocol frames** (Kogger SBP KP1/KP2, u-blox UBX, MAVLink 1/2). The module itself is a **Kogger SBP device** (board ID 87), configured and monitored by any SBP host from either port (X1 or X2) or over the network, found by KoggerApp's usual device search, and updated from a wire or the network with the exchange KoggerApp uses for Kogger devices. It reports what it sees on each port. It works as a Wi-Fi **station** or runs its own **access point** with a DHCP server. An optional **IP bridge** (SLIP + NAT) gives a Linux host without Wi-Fi an IP route through the module.

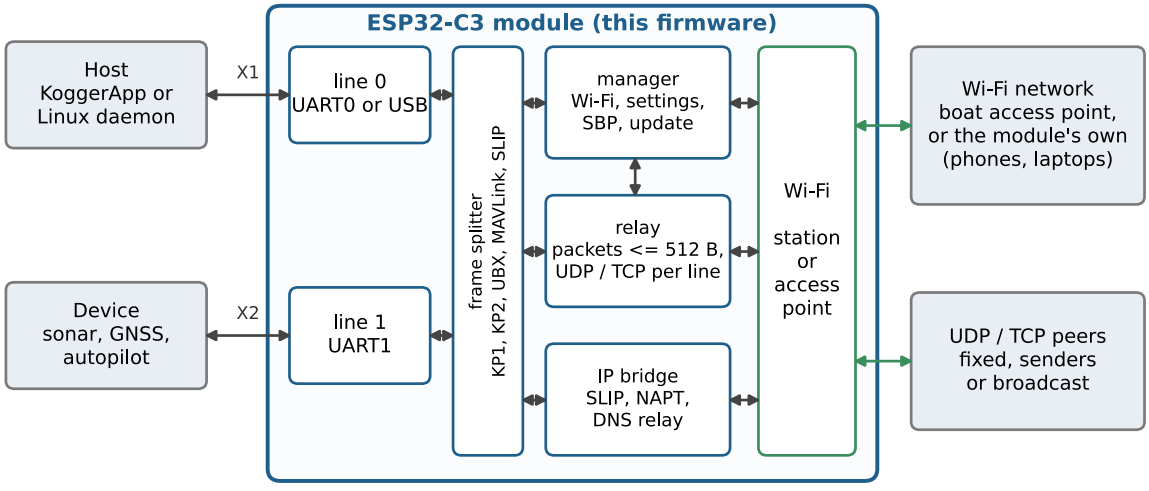
Features

- Frame-aware relay: frames up to 4096 B recognised by their own checksums; packets ≤ 512 B of whole frames; unknown data passes as raw bytes, in order
- Two lines, each: off, UDP (fixed peer, last ≤ 4 senders, broadcast) or TCP client
- Station (8 saved networks, auto-connect) or access point (≤ 10 clients, DHCP)
- Two equal ports: SBP control and relay on either, rate set per port, saved at once (BOOT 5 s: back to 921600)
- Answers discovery (`ID_VERSION` to address 0); replies go back on the request's channel
- Port information: which port the host is on, what is connected to each, devices heard, traffic by protocol
- Radio control: power 2–20 dBm, b/g/n, 20/40 MHz, LR, power save
- Safe update from a wire or the network (0.17): A/B slots, SHA-256, automatic rollback
- Link report with Wi-Fi traffic and the last reset reason

Example application

A boat head unit, a Linux display computer that talks Kogger SBP, has no usable Wi-Fi. The module sits on one of its UARTs at 3 Mbaud, joins the boat's access point and relays the SBP, MAVLink and UBX traffic between the head unit and the boat's equipment over UDP. On the same port the module answers its own SBP address 87 and reports its Wi-Fi state.

In the access-point role the same board is the boat-side end: it bridges a sonar or autopilot UART to phones and laptops on its own network.



Frames with the module's SBP address reach the manager from any line or from the network; everything else is relayed as whole frames. SBP and the IP bridge share line 0: the first frame after boot selects the protocol.

Figure 1. Architecture. Frames addressed to the module reach the manager; everything else is relayed.

Key specifications

Module	ESP32-C3-MINI-1U: RISC-V single core up to 160 MHz, 4 MB flash, U.FL antenna connector
Wi-Fi	IEEE 802.11 b/g/n, 2.4 GHz, 20/40 MHz; Espressif LR mode optional
Transmit power	2–20 dBm limit, applied by the driver in 11 steps
Roles	station (default) or access point: open, WPA2, WPA2/WPA3; channels 1–11; 1–10 clients
Serial lines	X1 = UART0 (host port), X2 = UART1; 9600–5 000 000 baud, 8N1; default 921600 on both; optional native USB Serial/JTAG build for line 0
Serial throughput	baud / 10 bytes per second each way: 92 KB/s at 921600, 200 KB/s at 2 Mbaud, 400 KB/s at 4 Mbaud
Relay	KP1, KP2, UBX, MAVLink 1/2 (392 messages); frames ≤ 4096 B; packets ≤ 512 B; 24-packet pool
Control protocol	Kogger SBP (KP1): `ID_WIFI` 0x57, `ID_WIFI_NET` 0x58, `ID_WIFI_SURVEY` 0x59, `ID_WIFI_RF` 0x5A, `ID_WIFI_PROFILE` 0x5B, common device IDs; IP bridge text commands over SLIP
Firmware update	over SBP, from a wire or the network (0.17); two 1.875 MiB slots; confirmation after 60 s, rollback at 180 s
Firmware image	0.95 MB (49 % of a slot), ESP-IDF 5.5.5
Board	20 × 30 mm, 4 layers FR4 1.2 mm, ENIG; two JST GH 4-pin connectors; KiCad 10 project in `hardware/`
Supply	4.5–24 V on X1 pin 4 (reverse-polarity diode, TI LMZM23601 3.3 V / 1 A module); consumption not measured

Board

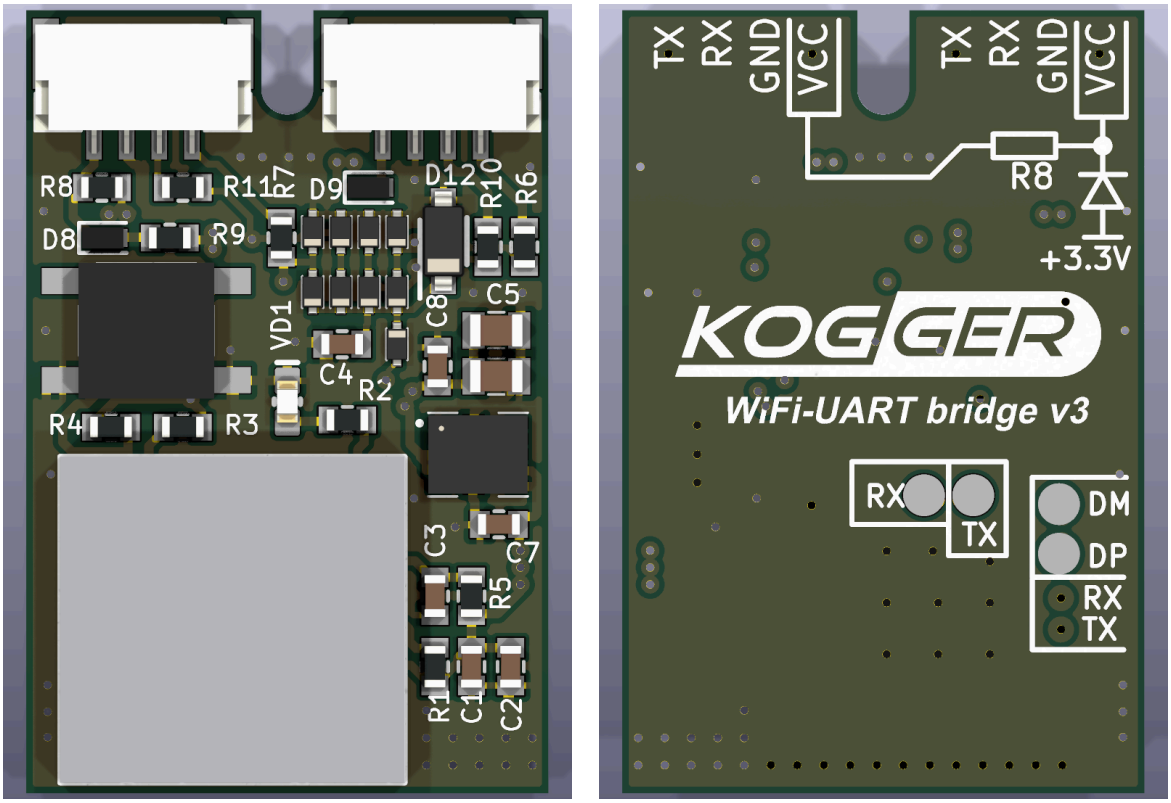


Figure 2. The reference board, 20 × 30 mm, top and bottom (3D render of the KiCad project). The KOGGER logo on the bottom is not licensed under the MIT License.

Connectors (JST GH, 1.25 mm, SM04B-GHS-TB; mating housing GHR-04V-S)

Pin	X1 (line 0, host port, flashing)	X2 (line 1)
1	TX of the module, 150 Ω series — GPIO21 (U0TXD)	TX of the module, 150 Ω series — GPIO5
2	RX of the module, 1 kΩ series — GPIO20 (U0RXD)	RX of the module, 1 kΩ series — GPIO4
3	GND	GND
4	VCC input through the Schottky diode D9 (reverse-polarity protection)	VCC_X2: VCC through R8 (0 Ω, fitted), or ≈ 3 V from +3.3 V through R9 + D8 without R8

Power

Regulator	TI LMZM23601V3: 4–36 V in, 3.3 V out, up to 1 A
VCC range	4.5–24 V, rated (not measured): regulator minimum plus the diode drop; below the 25 V input capacitors
Consumption	not measured; plan for ESP32-C3 Wi-Fi transmit peaks

Protection and control

- UART lines: series resistors, Schottky clamps to GND and to a Zener-held 3.3 V clamp rail.
- BOOT button on GPIO9; 10 kΩ pull-ups on GPIO2/8/9; EN 10 kΩ + 1 µF. No reset button, no auto-reset.
- Green LED on GPIO0 (active high): lit steadily without a link, a 100 ms flash every 2.1 s with one (0.20); blinks when a reset is armed. Bottom test pads: USB D−/D+, line 1 TX/RX.
- X2 is wired to GPIO5/GPIO4; the firmware can move line 1 to GPIO 0, 1, 3, 4, 5, 6, 7 or 10 on other boards.

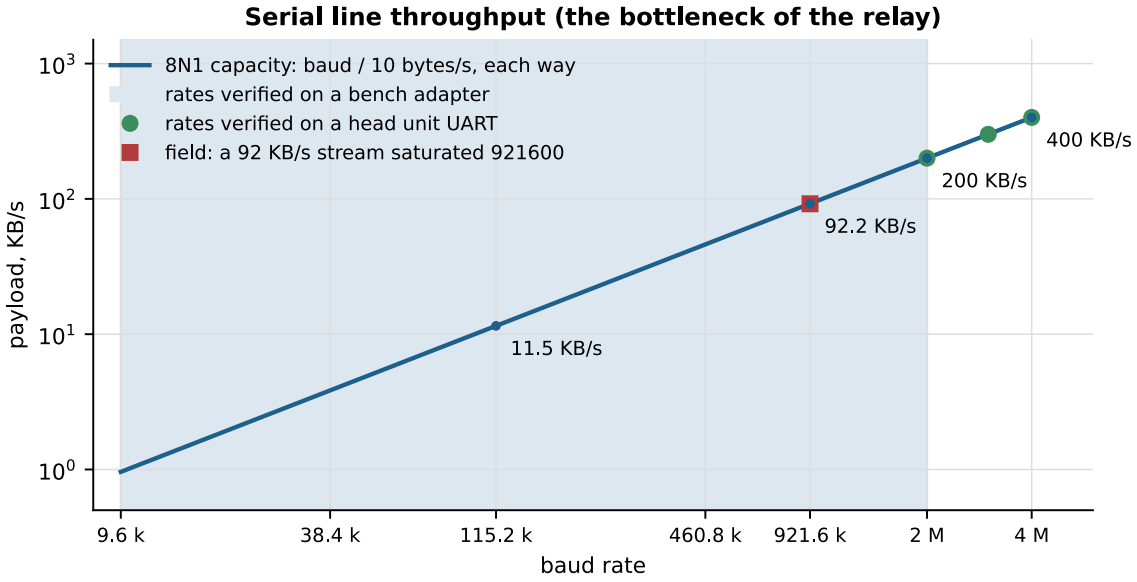


Figure 3. Serial throughput. At the supported rates the UART, not Wi-Fi, limits the relay.

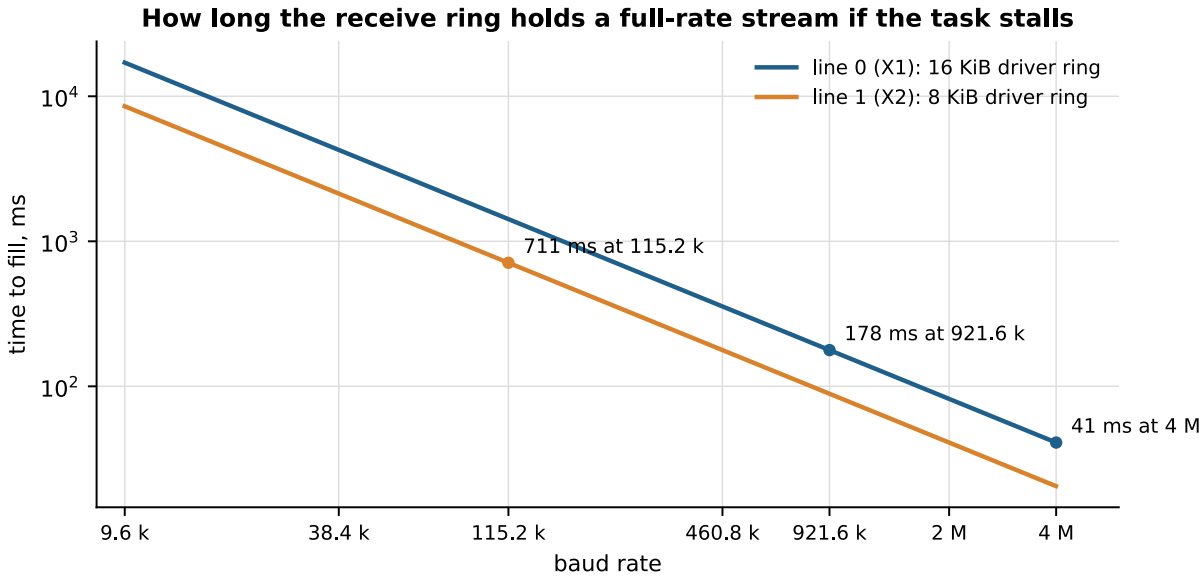


Figure 4. How long the UART driver's receive ring can hold a full-rate stream while the receive task is delayed.

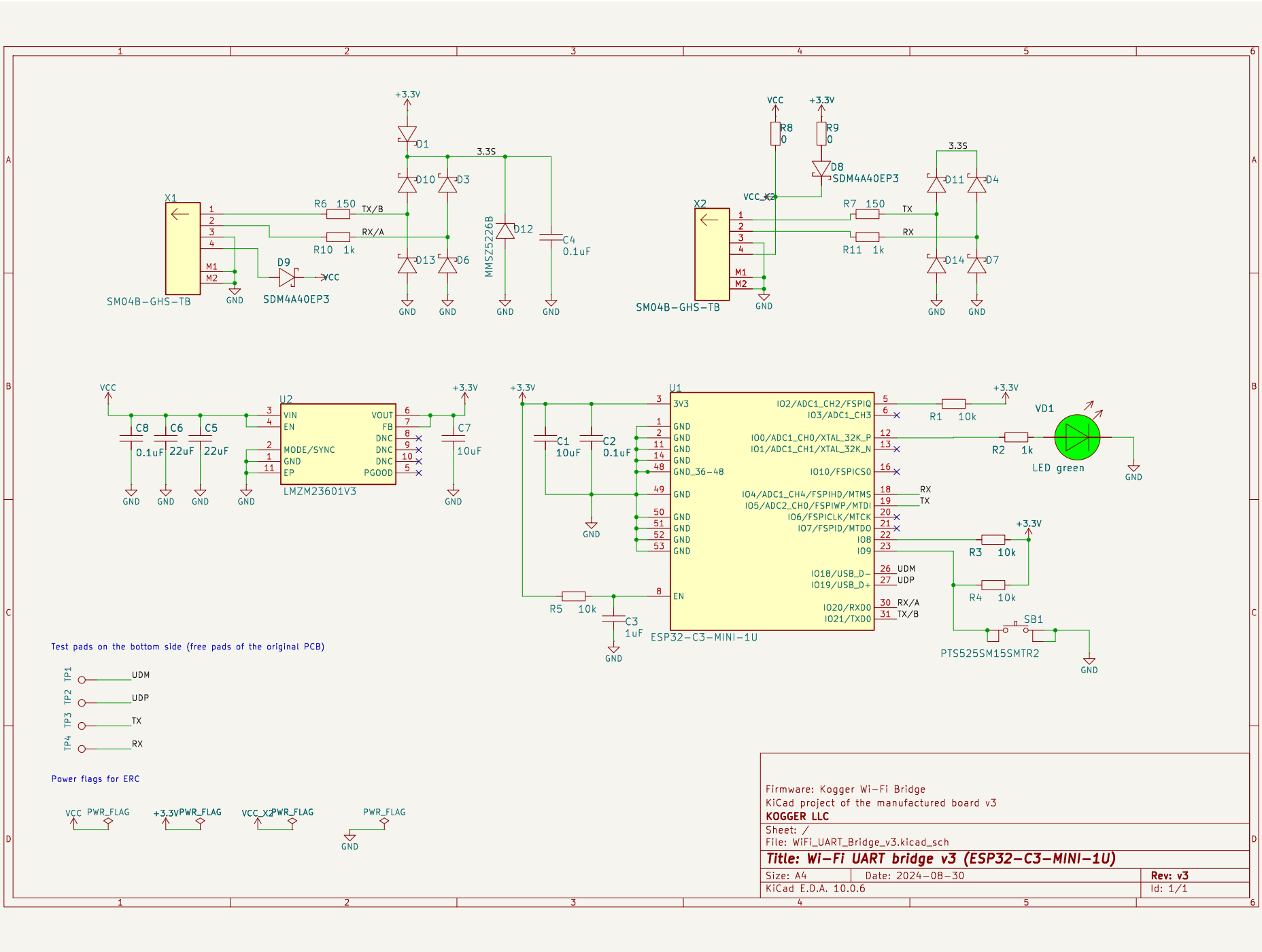


Figure 5. Schematic of the reference board (KiCad project in `hardware/`).

Bill of materials

References	Qty	Value	Manufacturer part number	Manufacturer
C1, C7	2	10uF	CL10A106MO8NQNC	Samsung Electro-Mechanics
C2, C4, C8	3	0.1uF	CL10B104JB8NNNC	Samsung Electro-Mechanics
C3	1	1uF	CL10A105KB8NNNC	Samsung Electro-Mechanics
C5, C6	2	22uF	CL21A226MAYNNNE	Samsung Electro-Mechanics
D1, D3, D4, D6, D7, D10, D11, D13, D14	9	BAS3005B-02V	BAS3005B-02VH6327	Infineon Technologies
D8, D9	2	SDM4A40EP3	SDM4A40EP3-7B	Diodes Incorporated
D12	1	MMSZ5226B	SZMMSZ5226BT1G	onsemi
R1, R3, R4, R5	4	10k	ERJ-3EKF1002V	Panasonic
R2, R10, R11	3	1k	ERJ-3EKF1001V	Panasonic
R6, R7	2	150	ERJ-3EKF1500V	Panasonic
R8, R9	2	0	ERJ-3GEY0R00V	Panasonic
SB1	1	PTS525SM15SMTR2	PTS525SM15SMTR2 LFS	C&K
U1	1	ESP32-C3-MINI-1U	ESP32-C3-MINI-1U	Espressif Systems
U2	1	LMZM23601V3	LMZM23601V3SILR	Texas Instruments
VD1	1	LED green	FYLS-0603UGC	—
X1, X2	2	SM04B-GHS-TB	SM04B-GHS-TB(LF)(SN)	JST

Test pads TP1–TP4 are not assembled. Fabrication: 4 layers, FR4 1.2 mm, 18 µm copper, ENIG, green solder mask, white silkscreen; Gerbers, drill, pick-and-place: `hardware/outputs/`.

Wi-Fi and network

Station (default role)

- Up to 8 saved networks, factory `KoggerBridge`; passwords are never read back.
- Scans every 15 s while searching, joins the strongest known network.
- 3 retries after a lost link, then search again; stuck states are supervised (30 s connect, 15 s scan).
- DHCP hostname `headunit-wifi` (build option).

Access point

- Name, password, security, channel 1–11, hidden flag, 1–10 clients: one SBP frame.
- DHCP server: pool ≤ 100 addresses, lease 1–2880 min, gateway/DNS optional.
- A refused configuration never leaves an open AP: previous settings, then defaults, else radio off.

Default	Station role	Access-point role
Line 0 (X1)	UDP to 10.0.0.10:14444	UDP, senders, port 14444
Line 1 (X2)	UDP to 10.0.0.10:14445	UDP, senders, port 14445
Module SBP address while a line relays	87	88
Wi-Fi network	`KoggerBridge` / `KoggerBridge` (public: change it): joined as a station, own network as an access point; WPA2, channel 6, 4 clients	
Address and DHCP	10.0.0.10/24, pool 10.0.0.11–10.0.0.30, lease 120 min, no gateway or DNS offered	
Radio	20 dBm, b/g/n, 20 MHz, no power save	

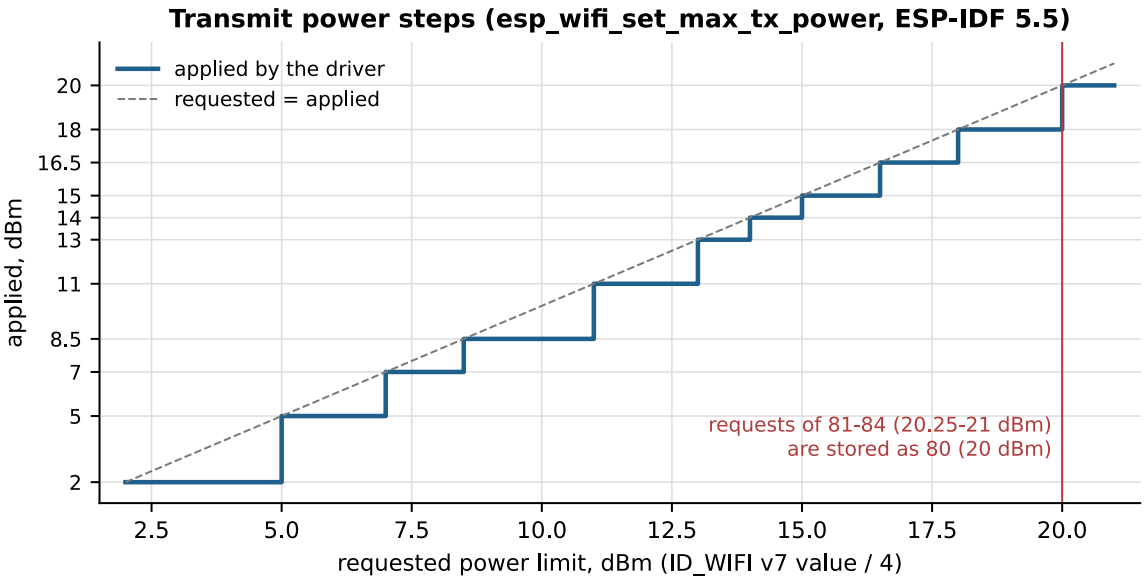


Figure 6. Requested and applied transmit power (ESP-IDF table). The module reports both.

Protocol reference

The module is controlled over **Kogger SBP** on either serial port (8N1, 9600–5 000 000 baud; 921600 on both by default), in either role, and over the network on the UDP port of any open line. X1 also understands SLIP frames of the IP bridge; the first valid frame after boot selects SBP or SLIP on X1 until the next reboot, and while line 0 relays to the network, X1 speaks only SBP. All multi-byte fields are little-endian. Full text: `docs/SBP_WIFI.md`.

Frame (KP1)

Offset	Field
0–1	sync `BB 55`
2	route: device address. The module answers its own address: 87 (station) / 88 (AP), or the one set with `ID_WIFI_NET` v6 (1–254). Discovery: a GETTING `ID_VERSION` to 0 or 255 is answered from the own address and relayed as well
3	mode = type | version<<3 | mark<<6 | resp<<7; type 1 CONTENT, 2 SETTING, 3 GETTING; version 0–7
4	id: command (0x57 `ID_WIFI`, 0x58 `ID_WIFI_NET`, common IDs below)
5	len: payload length, 0–255
6…	payload
end	ck1 ck2: Fletcher-8 over route … payload: `c1 = c2 = 0; for b: c1 = (c1 + b) & 255; c2 = (c2 + c1) & 255`

- **GETTING** vN → CONTENT vN (exception: `ID_WIFI` v3 → CONTENT v0). **SETTING** (sent with resp = 1) → acknowledgement CONTENT vN, resp = 1, payload `{code, ck1, ck2}`, where ck1 ck2 is the checksum of the acknowledged frame. Codes: 1 OK, 3 ERR_PAYLOAD, 5 ERR_VERSION, 6 ERR_TYPE, 7 ERR_KEY, 8 ERR_RUNTIME.
- Settings that can cut the module off start with the key `U4 0xC96B5D4A` (bytes `4A 5D 6B C9`). It guards against accidents, not attackers.
- CONTENT may grow in later firmware (fields are appended): parse by minimum length. Unknown IDs are ignored.
- Replies go back on the channel of the request. Unsolicited frames (state, periodic report, v7 after a connection, update progress) go to every channel that asked the module something within the last 60 s; a port with only a device on it gets nothing.

Common device IDs

ID	GETTING	SETTING
`ID_UART` 0x18	v0 {KEY, 1, U4 baud of the asking port}; v1 {KEY, 1, own address}; v2 {KEY, own address}	v0 {KEY, U1 1, U4 baud 9600–5 000 000} for the asking port: answered at the old rate, then switched and saved (0.15; provisional up to 0.14); v1/v2 own address 1–254, saved
`ID_VERSION` 0x20	v0 34 bytes ([1] board 87, [14..17] serial); v1 12-byte UID; v2 9 bytes {bootMode, 0, 87, 0, 0, U2 0, fwMinor, fwMajor}	—
`ID_MARK` 0x21	v0 {U1 mark}	v0 {KEY}: sets the mark bit (cleared by a reboot and by `ID_BOOT` v0)
`ID_FLASH` 0x23	—	{KEY}: v0 confirm and save the rates of both ports and the report period; v1 reload the saved period; v2 erase
`ID_BOOT` 0x24	—	v0 {KEY}: reboot after the 5 s update window; v1 {KEY}: check the received image and boot it
`ID_UPDATE` 0x25	—	v0 {U2 chunk number, ≤ 253 data bytes}; from a wire or the network (0.17)

`ID_WIFI` = 0x57

Ver.	GETTING → CONTENT	SETTING
v0	state, 39 bytes + SSID (below); also sent by itself on every change	— (ERR_TYPE)
v1	link report, 27 bytes: U1 state, S1 RSSI, U4 rx B/s, U4 tx B/s, U4 rx total, U4 tx total, U4 uptime s, U1 last reset reason (esp_reset_reason_t: 1 power-on, 3 software, 4 panic, 6 task watchdog, 9 brownout), U4 requests dropped for a full queue (0.14, stays 0); sent periodically	{U2 period ms}: 0 off, 100–60000
v2	scan: one CONTENT per network {U1 index, U1 total, S1 RSSI, U1 channel, U1 security, U1 flags, BSSID[6], U1 len, SSID}; none: {0, 0}	start a scan (OK); ERR_RUNTIME while connecting or in the AP role
v3	state: answered with CONTENT v0	connect {U1 flags bit0 save, U1 len, SSID 1–32, U1 len, password 0–64}
v4	saved networks {U1 index, U1 total, U1 len, SSID}; passwords never	{0} disconnect, auto off; {1} auto on; {2, U1 len, SSID} forget one; {3} forget all
v5	relay of line 0, 10 bytes	{U1 mode 0 off/1 UDP/2 TCP, U1 own address 1–254, ip[4], U2 peer port, U2 own port (0 = peer port)}; ip 0.0.0.0 answers senders, 255.255.255.255 broadcasts
v6	relay statistics, 47 bytes: U1 state, ip[4], U2 port, U4 ×10 (up frames, bytes, packets, drops; down packets, bytes, frames, drops; frames for the module; TCP connects)	— (ERR_TYPE)
v7	radio, 8 bytes: the 4 settings, then power in effect, negotiated mode (0 LR, 1 11b, 2 11g, 4 HT20, 5 HT40, 0xFF none), protocols and bandwidth set in the driver	{U1 power 0.25 dBm 8–80, U1 protocols 0x01 b/0x03 b/g/0x07 b/g/n/ 0x08 LR/0x0F b/g/n+LR, U1 bandwidth 1 = 20/2 = 40 MHz, U1 power save 0/1/2}

Offset	CONTENT v0 (state)
0	U1 state: 0 IDLE, 1 SEARCHING, 2 CONNECTING, 3 CONNECTED, 4 AP
1	U1 flags: bit0 auto-connect, bit1 scan running, bit2 current network saved, bit3 radio switched off (0.20)
2 / 3 / 4	S1 RSSI dBm / U1 channel / U1 security (0 open, 1 WEP, 2 WPA, 3 WPA2, 4 WPA/WPA2, 5 WPA3, 6 WPA2/WPA3, 7 enterprise, 8 OWE)
5 / 6 / 7	U1 last disconnect reason (wifi_err_reason_t) / U1 category (0 none, 1 no AP, 2 auth, 3 assoc, 4 lost, 5 other) / U1 AP clients
8–13	BSSID
14–29	IP, netmask, gateway, DNS (4 bytes each)
30–37	U4 receive, U4 transmit traffic, bytes per second
38	U1 SSID length, then the SSID (up to 32 bytes)

`ID_WIFI_SURVEY` = 0x59 (0.16)

How busy each channel is, to put an access point on a quiet one. `SETTING v0 {KEY, U2 dwell ms, U2 channel mask}` sweeps the channels (1–11, 50–1000 ms each, 120 by default) and answers OK at once; `GETTING v0` repeats the last results. `CONTENT v0`, one per channel, 16 bytes: index, total, channel, dwell, **busy per mille** (air time of the Wi-Fi frames heard, without gaps or acknowledgements: a floor), frames, transmitters, strongest RSSI, noise floor, flags (own channel, usable by an access point, noise not to be trusted) and how much of the busy figure is the module's own network. `GETTING v1 {U2 window ms}` (0.18) reports the **link rate**: the PHY rates the other end of the link transmits at, counted over a 200 ms sniffer window, at most once a second (11b, 11g and 11n decoded to kbit/s). Since 0.19 every rate carries a timing bound from consecutive frames, which proves an LR rate of 500 kbit/s when it reaches 300 kbit/s (LR has only 250 and 500). Since 0.20 a page ends with how many of those frames were retransmissions (the Retry bit): the share of the other end's retransmissions.

`ID_WIFI_RF` = 0x5A (0.20)

The radio switched off, to swap an antenna while the module runs. `SETTING v0 {KEY, U1 on, U2 seconds}`: on = 0 stops the whole radio for that many seconds (0 = until an "on" request or a reboot), on = 1 starts it again; nothing is saved. A request that came over the network must carry a timer of 1–600 s (ERR_PAYLOAD otherwise): the module is reached through that very radio.

The answer leaves before the radio stops. `CONTENT v0 {U1 on, U2 seconds left}` answers GETTING and SETTING. While off: `ID_WIFI` v0 IDLE with flag bit3, the LED lit steadily, and scans, joins, sweeps and access point changes refused with ERR_RUNTIME.

`ID_WIFI_PROFILE` = 0x5B (0.22)

Every saved setting as one profile, so a module is never left half set up. `SETTING v1 {KEY, U2 length, U4 CRC-32}` begins an upload of up to 1024 bytes, `SETTING v2 {U2 offset, bytes}` carries it in order in pieces of up to 253 bytes, `SETTING v3 {KEY}` applies it: the module checks the whole profile with the rules of the single settings and either refuses (ERR_PAYLOAD and `CONTENT v3 {U1 reason}`, nothing saved) or saves everything at once, answers OK and reboots 200 ms later into the new configuration. `GETTING v0` reports the profile's name and whether a setting changed since. The blob: name, role, own address, X1 rate, radio, access point, address and DHCP, both lines (line 1 with its rate, pins and input translator) and up to 8 station networks. Passwords are never read back.

`ID_WIFI_NET` = 0x58 (every SETTING starts with the key)

Ver.	GETTING → CONTENT	SETTING (after the key)
v0	role {U1 at next boot, U1 now}: 0 station, 1 AP	{U1 role, U1 flags: bit0 reboot now, bit1 forget lines and own address}
v1	access point {U1 channel 1–11, U1 hidden, U1 max clients 1–10, U1 security 0 open/1 WPA2/2 WPA2+WPA3, U1 len, SSID, U1 0}	the same + {U1 len, password 8–63}; empty password with security on keeps the old one
v2	address, 20 bytes {ip[4], mask[4], U1 DHCP, pool start[4], pool end[4], U2 lease min, U1 offer bit0 gateway, bit1 DNS}	the same 20 bytes
v3	line, 18 bytes {U1 line, U1 mode 0 off/1 UDP/2 TCP, U1 dest 0 fixed/1 senders/2 broadcast, ip[4], U2 rport, U2 lport, U4 baud (running), S1 TX pin, S1 RX pin, U1 uart}, from 0.21 21 bytes: + {U1 input translator saved, U1 running, U1 translators built}; no payload: both lines	the first 17 bytes; baud sets the other line (either from the network), saved at once; ignored for the asking port's own line; pins to line 1 only; an 18th byte (0.21): line 1's input translator, from the next boot
v4	line statistics, 57 bytes {U1 line, U1 state, ip[4], U2 peer port, U1 senders, U4 ×12: up frames, bytes, packets, drops; down packets, bytes, frames, drops; frames for the module; TCP connects; UART RX overflows; UART TX drops}	—
v5	AP clients {U1 index, U1 total, MAC[6], ip[4], S1 RSSI} each; none {0, 0}	—
v6	own address {U1}	{U1 address 1–254}, acknowledged from the old address
v7	port information: {} both ports, {port}, {port, page}. Page 0, 116 bytes: flags (UART, bridging, **asked through this port**, gets reports, SLIP, provisional up to 0.14, USB), rate now and saved, connected (nothing, unreadable, SBP host, SBP devices, both, MAVLink, u-blox, IP bridge), bytes, units by protocol both ways, SBP requests/CONTENT, overflows. Page 1: devices heard (SBP address, board, firmware, serial; MAVLink system, type, autopilot). Page 2: network peers of the line	— (ERR_TYPE)

Firmware update exchange

Step	Host sends	Module answers
1	`ID_BOOT` v0 {KEY}	OK; update window 5 s; `ID_VERSION` v2 bootMode = 1; no chunk → reboot
2	`ID_UPDATE` v0 {U2 n, data} for n = 1, 2, …	CONTENT {U2 lastNumMsg, U4 lastOffset, U1 type, U1 rcvNumMsg}: type 0 accepted, 2 reposition to lastNumMsg + 1, 3 no session, 4 foreign image, 5 too big, 6 flash error
3	`ID_BOOT` v1 {KEY}	OK and reboot if the image checks out (chip, project, SHA-256); ERR_RUNTIME if it does not or the transfer was abandoned; OK without reboot if nothing was received
4	any frame to the module	the new image confirms itself after ≥ 60 s of uptime with host frames; else rollback at 180 s

Worked examples

Generated with `host/sbpframe.py`; `tools/make_datasheet.py` re-checks every checksum below.

What	Bytes (hex)
Find the module (discovery): GETTING ID_VERSION v2 to address 0	`BB 55 00 13 20 00 33 79` answered from the module's own address (87 station, 88 AP) and relayed on to a device behind it; KoggerApp sends this when a port opens
Read identity: GETTING ID_VERSION v0	`BB 55 57 03 20 00 7A A5` answer: CONTENT v0, 34 bytes, board 87 at [1]
Read Wi-Fi state: GETTING ID_WIFI v0	`BB 55 57 03 57 00 B1 13` answer: CONTENT v0 (39 bytes + SSID)
Which port am I on, what is on each: GETTING ID_WIFI_NET v7	`BB 55 57 3B 58 00 EA BD` answer: CONTENT v7 page 0 (116 bytes) for X1 and for X2; flags bit2 marks the port the request came through
Report every 500 ms: SETTING ID_WIFI v1 {U2 500}	`BB 55 57 8A 57 02 F4 01 2F 07`
its acknowledgement: CONTENT v1 resp {OK, ck1, ck2}	`BB 55 57 89 57 03 01 2F 07 71 BE` ck1 ck2 = checksum of the SETTING frame above
This port to 2 Mbaud: SETTING ID_UART v0 {KEY, 1, U4 2000000}	`BB 55 57 82 18 09 4A 5D 6B C9 01 80 84 1E 00 F8 D7` acknowledged at the old rate, then switched; saved when the next request arrives at 2 Mbaud, otherwise back after 10 s
X2 to 57600 from X1: SETTING ID_WIFI_NET v3 (line 1 record)	`BB 55 57 9A 58 15 4A 5D 6B C9 01 01 01 00 00 00 00 6D 38 6D 38 00 E 1 00 00 05 04 70 DB` the other port: saved at once
Join a network: SETTING ID_WIFI v3 {save, "Boat-AP", "secret12"}	`BB 55 57 9A 57 12 01 07 42 6F 61 74 2D 41 50 08 73 65 63 72 65 74 3 1 32 97 54` OK at once, progress in CONTENT v0
Radio: SETTING ID_WIFI v7 {15 dBm, b/g/n, 20 MHz, no power save}	`BB 55 57 BA 57 04 3C 07 01 00 B0 F3`
Relay line 0 over UDP to 10.0.0.10:14444, own address 87: SETTING ID_WIFI v5	`BB 55 57 AA 57 0A 01 57 0A 00 00 0A 6C 38 6C 38 16 E9` the address 87 is saved
AP name "Boat-1", WPA2, channel 6, 4 clients, password kept: SETTING ID_WIFI_NET v1	`BB 55 57 8A 58 10 4A 5D 6B C9 06 00 04 01 06 42 6F 61 74 2D 31 00 1 9 1B`
Role access point, reboot now: SETTING ID_WIFI_NET v0	`BB 55 57 82 58 06 4A 5D 6B C9 01 01 14 79` after the reboot the module answers its saved address, here 87 (saved with ID_WIFI v5 above); with nothing saved, or with flags 0x03 (forget lines and address), the access point default 88
Periodic report from the module: CONTENT ID_WIFI v1 (23 bytes)	`BB 55 57 09 57 17 03 CC DB 16 00 00 62 07 00 00 E0 FD 1C 00 40 EF 0 7 00 10 0E 00 00 01 45 0E` CONNECTED, RSSI -52 dBm, 5851/1890 B/s, totals, uptime 3600 s, last reset: power-on
Radio off for 60 s (antenna swap): SETTING ID_WIFI_RF v0 {KEY, 0, U2 60}	`BB 55 57 82 5A 07 4A 5D 6B C9 00 3C 00 51 1A` answer: OK, then CONTENT v0 {0, U2 60}; the radio stops 200 ms later and comes back by itself
Radio on now: SETTING ID_WIFI_RF v0 {KEY, 1, 0}	`BB 55 57 82 5A 07 4A 5D 6B C9 01 00 00 16 A5`
Which profile the module carries: GETTING ID_WIFI_PROFILE v0	`BB 55 57 03 5B 00 B5 1B` answer: CONTENT v0 {flags: bit0 loaded, bit1 changed since; U4 fingerprint; U2, U2 upload in progress; name}
Apply an uploaded profile: SETTING ID_WIFI_PROFILE v3 {KEY}	`BB 55 57 9A 5B 04 4A 5D 6B C9 2B 02` answer: OK, CONTENT v3 {0} and CONTENT v0, then the reboot; or ERR_PAYLOAD and CONTENT v3 {reason}, nothing changed
Update, step 1: SETTING ID_BOOT v0 {KEY}	`BB 55 57 82 24 04 4A 5D 6B C9 DC 10` opens the 5 s update window
Update, step 2: SETTING ID_UPDATE v0 {U2 1, 4 data bytes}	`BB 55 57 82 25 06 01 00 E9 05 02 2F 24 36` chunks up to 253 data bytes, numbered from 1; the first bytes of an ESP-IDF image are E9 ..

Update, step 3: SETTING ID_BOOT v1 {KEY}	`BB 55 57 8A 24 04 4A 5D 6B C9 E4 48` checks the image, answers OK and reboots

IP bridge (SLIP) and text commands

Frame `C0 | SLIP-escaped(type, data, CRC-16/CCITT-FALSE big-endian) | C0`; type 0x01 carries one IPv4 packet, type 0x02 one text line `<tag> <COMMAND> [key=value …]`. Commands: VER, STATUS, SCAN, CONNECT, DISCONNECT, AUTO, LIST, FORGET, STATS, BAUD, PROTO, RADIO, RELAY, REBOOT; events start with `*`. Example, text `1 VER`: `C0 02 31 20 56 45 52 70 41 C0`. Details: `docs/PROTOCOL.md`.

Relay

Frame	Start	Check
Kogger KP1	`BB 55 route mode id len`	Fletcher-8
Kogger KP2	`CC 55 U2(total)`, ≤ 4096 B	Fletcher-8
u-blox UBX	`B5 62 class id U2(len)`, ≤ 4096 B	Fletcher-8
MAVLink 1 / 2	`FE` / `FD`, known message id and length	X.25 + `CRC_EXTRA`

- A frame is emitted only after its checksum matches; after a false candidate the search resumes at the next byte.
- A packet leaves when full, when the UART pauses, or after 10–20 ms (UART side) / 200 ms (network side) of silence.
- Overload drops whole units, never fragments; the module's own frames keep a 1 KB reserve in each transmit ring.
- A rescan budget bounds the work on hostile input (Figure 7).

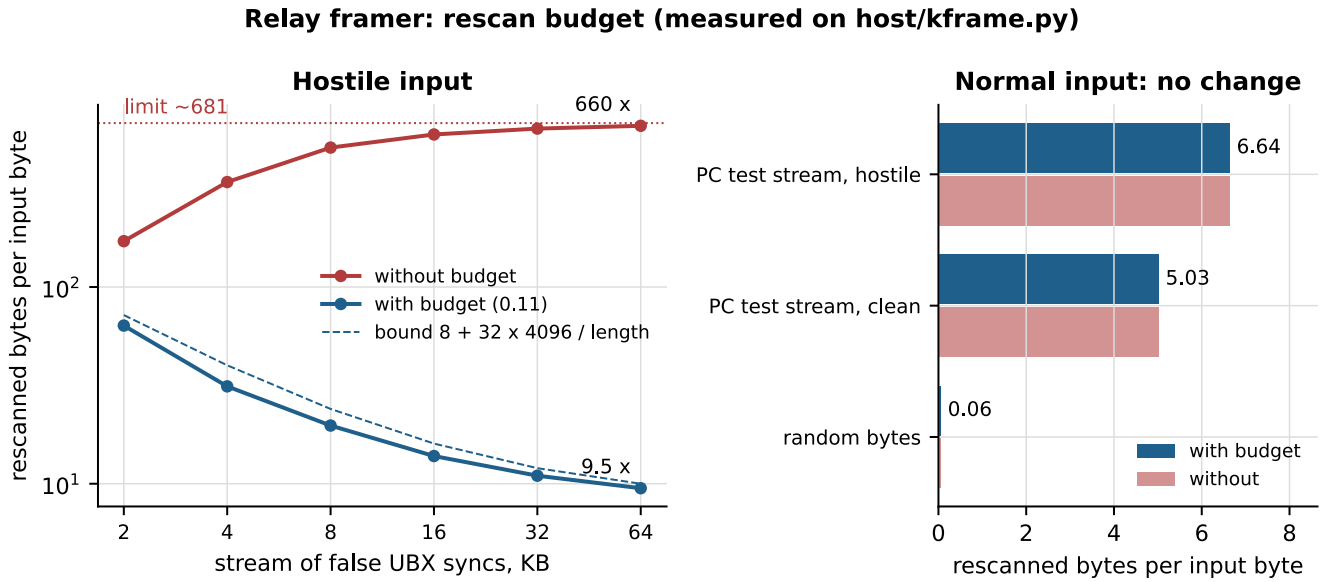


Figure 7. Rescanned bytes per input byte, measured on the Python mirror of the framer (byte-equal to the firmware's C in the PC tests).

Firmware update

`ID_BOOT v0` → `ID_UPDATE` chunks (≤ 253 B, strictly numbered) → `ID_BOOT v1`, the exchange KoggerApp uses for Kogger bootloaders. Only the inactive slot is written; the image (chip, project, SHA-256) is checked before the switch. The new image confirms itself after 60 s of work with host frames; otherwise the bootloader rolls back after 180 s. Updates from the network are refused. File: `KoggerWiFi_<version>.ufww` (`tools/make_update.py`); host tool: `tools/sbp_update.py`.

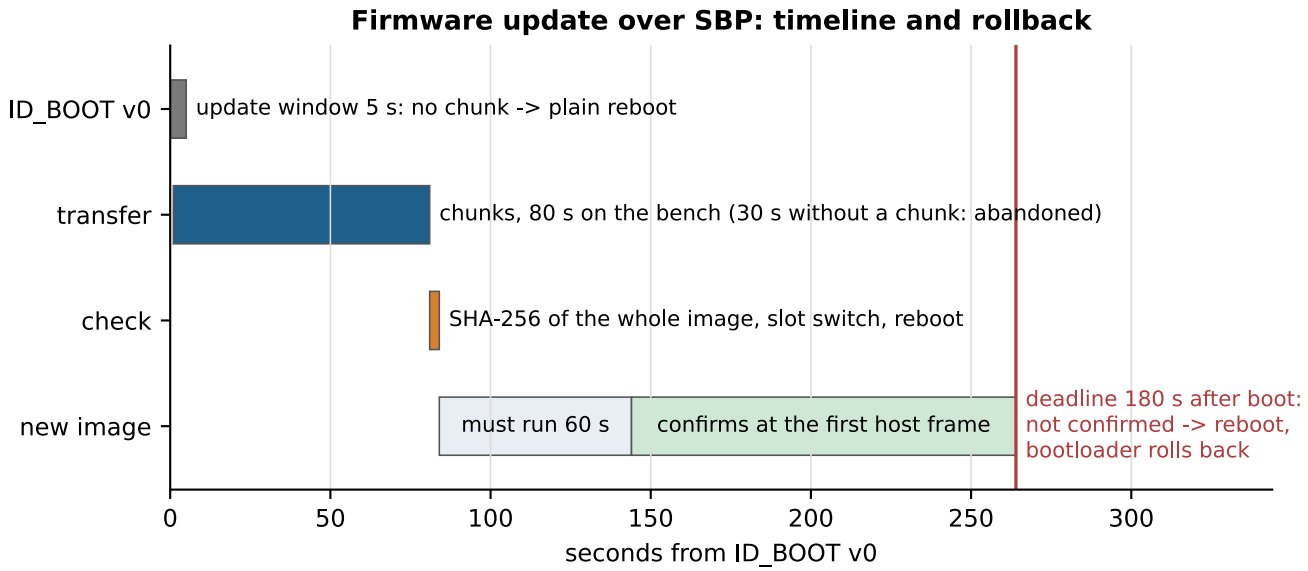


Figure 8. Update timeline with the bench transfer time (885 KB in 80 s at 921600 baud).

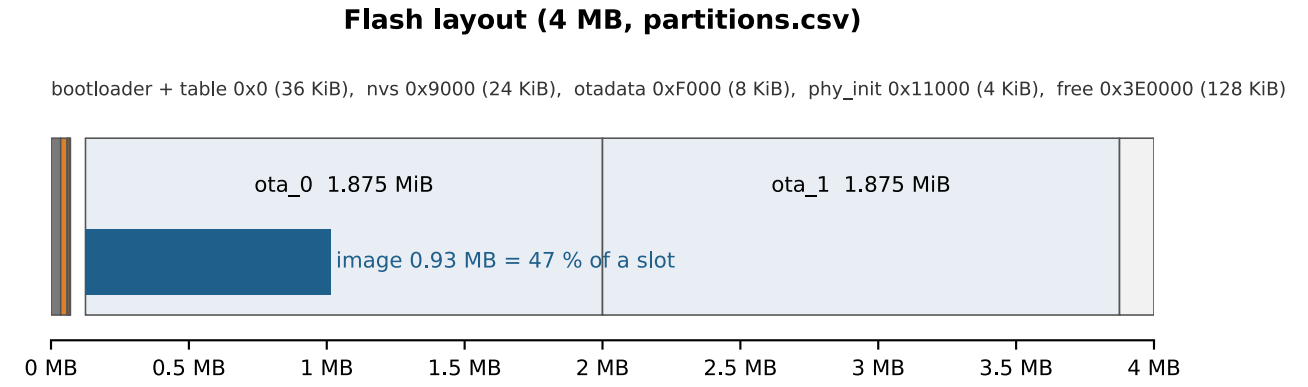


Figure 9. Flash layout from `partitions.csv`.

Parameters

Timing

Update window	5 s after `ID_BOOT` v0
Update stall	30 s
Confirmation	≥ 60 s uptime + host frame, deadline 180 s
Network search	every 15 s
Supervision	connect 30 s, scan 15 s, lost association 2 s
Relay silence flush	10–20 ms UART, 200 ms network
Relay senders	≤ 4, forgotten after 60 s
TCP	connect 3 s, send 5 s, retry 2 s
Link report	1000 ms (100–60000, or off)
Task watchdog	5 s, panic → reboot
BOOT resets	held 5 s: port rates; 10 s: factory settings

Buffers

Line 0 receive	16 KiB driver ring
Line 0 transmit	12 KiB whole-unit ring (1 KiB reserve), no driver ring
Line 1 receive / transmit	8 KiB / 8 KiB (1 KiB reserve)
Packet pool	24 × 512 B, both lines
Frame candidate	4096 B per direction
lwIP UDP queue	16 datagrams
DHCP leases	10
Scan results	32 networks
Rescan budget	8 B per input byte, saved up to 128 KiB

Build options (menuconfig “HeadUnit Wi-Fi bridge”)

Option	Default	Meaning
`WB_LINK`	USB Serial/JTAG	host link; `sdkconfig.defaults.uart` selects UART0 at 921600
`WB_LINK_UART_TX / RX / RTS / CTS`	21 / 20 / −1 / −1	host link pins; −1 = no flow control
`WB_BRIDGE_IP / WB_HOST_IP`	10.99.0.1 / 10.99.0.2	IP bridge point-to-point addresses
`WB_OTA_CONFIRM_S`	180	rollback deadline, 90–3600 s
`WB_HOSTNAME`	headunit-wifi	DHCP hostname of the station
`WB_FACTORY_SSID / PASS`	KoggerBridge	factory network in both roles (public)

Measured performance

Test	Conditions	Result
Ping through the IP bridge and NAPT	bench, 921600 baud, 30 pings	56 B: 4.9 / 6.1 / 17.4 ms; 1400 B: 40.4 / 41.4 / 46.3 ms (min/avg/max), data intact
Firmware update	bench, 921600 baud	885 120 B in 80 s (10.8 KB/s); lost chunk recovered
Firmware update	field, saturated 921600 port	929 KB in 226 s, 5/5 checks
Firmware update 0.10 → 0.11	field, 3 Mbaud	932 512 B in 50 s (18.1 KB/s), 7/7 checks
Firmware update 0.11 → 0.12	bench, 921600, AP role	938 800 B in 84 s (10.9 KB/s), 7/7 checks
Relay to a boat network	field, UDP	~270 datagrams, ~890 frames in 8 s, no loss
Auto-connect after power-up	bench	connected after 7.6 s without commands
Baud switching	bench adapter / head unit	9600 … 2 000 000: 28/28 / 2, 3, 4 Mbaud

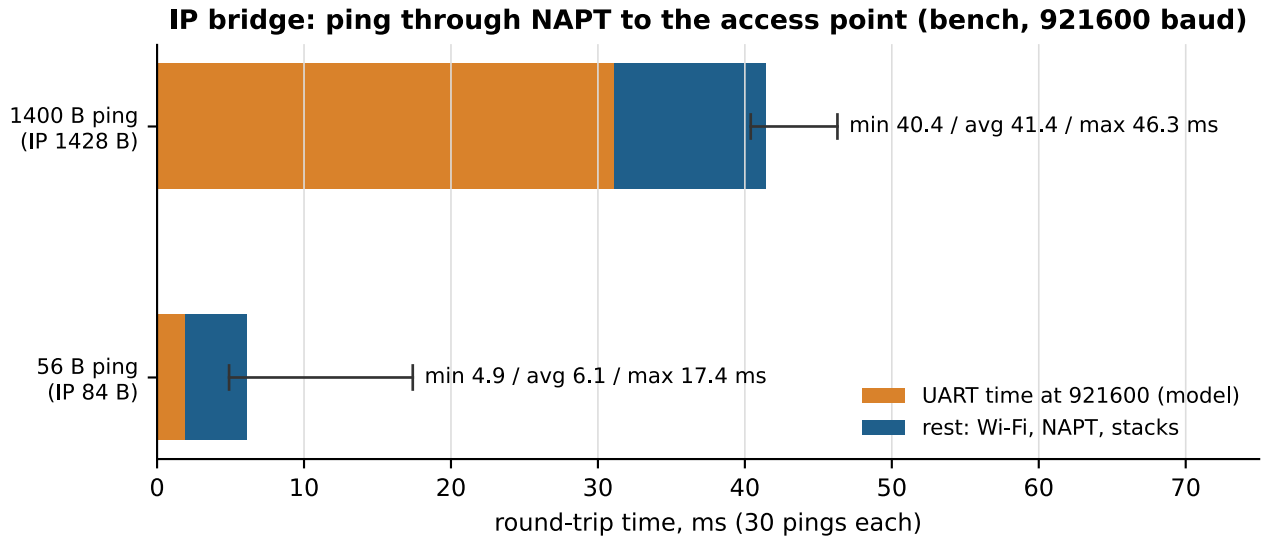


Figure 10. Round-trip time through the IP bridge, split into the UART share (model) and the rest.

Status and security

Verified on hardware (0.1–0.12): SBP device checks, baud switching, IP bridge through NAT, station auto-connect, firmware update, recovery from a lost chunk, a corrupted image refused, relay to a boat network over UDP; 0.11.0 installed in the field over SBP (7/7 checks); 0.12.0 port rules on a bench module in both roles (29/29); 0.12.0 on the head unit's module (7/7) with a sonar on the access point's X2 seen over an LR-only link. Not yet verified on hardware: the 0.11 transmit fix under a saturated 921600 port; rollback of an unconfirmed image and an abandoned transfer; a host on X2; the native USB variant; the IP bridge daemon on a Linux host; LR range and throughput over distance.

> **Security.** Kogger SBP has no authentication; any client of the module's network can reconfigure it. Images are not signed; updates are accepted only over a wired line. The default access-point password is public. Change it before using the access-point role and keep the module on a trusted network (`SECURITY.md`).

Documentation: README.md, docs/HARDWARE.md, docs/SBP_WIFI.md, docs/NETWORK.md, docs/RELAY.md, docs/UPDATE.md, docs/PROTOCOL.md, docs/PARAMETERS.md, docs/DESIGN.md, docs/BUILD_AND_TEST.md. Third-party notices: THIRD_PARTY.md. This datasheet is generated from docs/datasheet/datasheet.html by tools/make_datasheet.py.